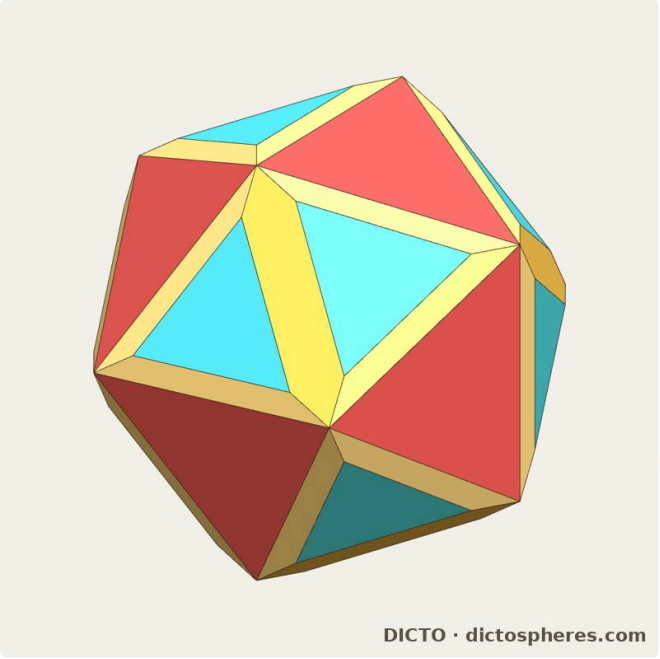
The DICTO OCD Stella

The Ogdoecontahedron's companion cut along its joints: a golden solid that fills 151/2 − 267√5/8 ≈ 87.12% of space in the Ogdoecontahedron's packing, copies meeting face to face

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Version 1 · 17 October 2026 · DICTO DOI: on publication · Licence CC BY 4.0 · An introduction to one solid; its parent paper: The DICTO Ogdoecontahedron, DICTO DOI: 10.5281/zenodo.23404688. Data and checks are included with this record.
The companion ("a free gift from the way the adapters sat on the OCD"), the stella cut, the seed and the name are DICTO's; the exact computations and the text were done with an AI assistant (Claude, Anthropic).

Abstract. The DICTO Ogdoecontahedron (OCD) packs in a body-centred cubic arrangement, each copy joined to 8 neighbours on hexagons. The golden adapter pieces around one copy have an icosahedral hull, the OCD's companion C (72 corners: 30 hexagons in golden planes and 20 triangles). Cutting C along the 8 joints gives the DICTO OCD Stella S: 48 corners and 50 faces, golden except for the joint triangles, which are exactly the packing cell's hexagon walls. Copies of S meet face to face across all 8 joints and fill exactly 151/2 − 267√5/8 ≈ 87.12% of space. Twelve OCD Stellas around one small one fill an outer OCD Stella exactly at φ^2, the Bodhi seed's pattern, though S is only pyritohedral. Exact in $\mathbb{Q}(\sqrt{5})$ unless marked float.



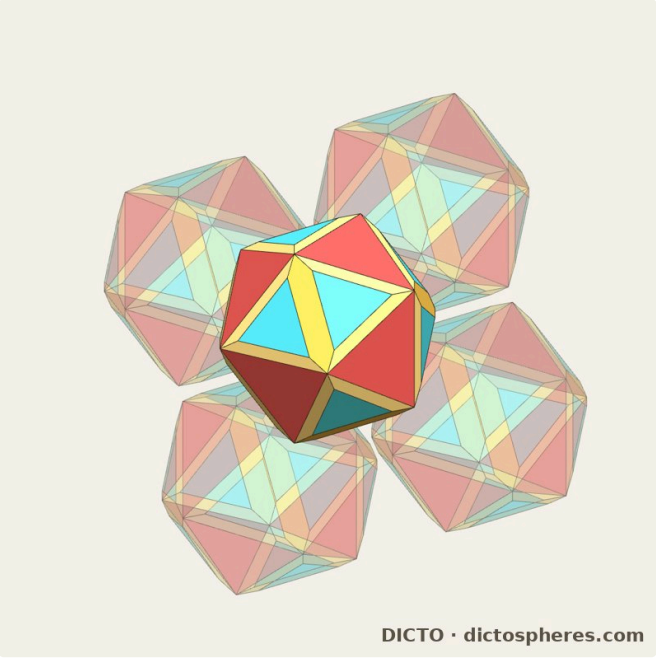


Figure 1. Left: the DICTO OCD Stella. Right: copies in the OCD's packing, meeting face to face across the joints.

1. The solid

Around one OCD of its packing (cell volume 512/729 + 256√5/729), the 60 golden adapter pieces (the full orbit of the 24 per cell) have a convex hull C with 72 corners, 30 hexagons in the golden planes of the OCD's envelope and 20 triangles in the three-fold planes through the adapters' outer corners (squared distance 14/27 − 10√5/243), volume (−18880 + 9760√5)/2187. Its combinatorial type is Conway t3jD, the triacontahedron truncated at its 20

three-fold corners, here at the depth set by the OCD (Antiprism). The DICTO OCD Stella is S = C cut by the packing's Voronoi cell, that is, along the 8 joints.

Property	Value (exact)
corners, faces	48 corners; 50 faces: 6 golden hexagons, 24 golden quadrilaterals, 12 triangles of C, 8 triangles in the OCD's joint hexagon planes (the cell's hexagon walls)
volume	$(-4064 + 2240\sqrt{5})/729 \approx 1.296011$
packing	fills $151/2 - 267\sqrt{5}/8 \approx 87.12\%$ of space in the OCD's packing; copies meet face to face across the 8 joints
symmetry	pyritohedral (the cut keeps the 24 symmetries the OCD shares with the cubic packing)

Table 1. The DICTO OCD Stella (`stella_cut_exact.py` → `stella-cut-exact.txt`).

2. The seed and the crystals

The seed (exact). An outer OCD Stella of scale φ^2 holds 12 OCD Stellas on its five-fold axes, each touching it, and one of scale $1/\varphi$ at the centre touching all 12. By the Minkowski argument the seed closes exactly when no two of the twelve overlap; of the 66 pairs, 30 have gauge exactly 1 (touching, each copy 5 others, along the icosahedron's edges) and 36 exactly φ. So it closes exactly at φ^2 with the centre at $1/\varphi$, and the gap is $-69 + 31\sqrt{5} \approx 31.8\%$: the pattern of the Bodhi Seed (DICTO DOI: 10.5281/zenodo.23311353), though the OCD Stella is only pyritohedral (`stella_seed_exact.py`).

Crystals. On one network of the OCD's packing with a partner on the other: with icosahedra in the OCD's 20 hexagon planes, every solid meets its 8 joint neighbours face to face and the fill is $1523/32 - 669\sqrt{5}/32 \approx 84.60\%$ (exact); with regular octahedra, $\approx 96.33\%$ (float); alone on both networks, 87.12% (exact). With icosahedra or octahedra the crystal has space group Fd-3 (No. 203).

3. Status

Known: the combinatorial type of C (Conway t3jD, the chamfered-icosahedron family). **Not yet searched comprehensively:** the OCD Stella itself, its packing and its seed. We state no priority claim here beyond "derived by DICTO from the Ogdoecontahedron's packing"; a prior-art search is to follow.

4. Files

In the study (branch research: studies/2026-10-16-ocd-pyrochlore): `stella_cut_exact.py` → `stella-cut-exact.txt`, `stella_seed_exact.py` → `stella-seed-exact.txt`, `ocd_gbpc_hull.py`, `companion_variants.py`. See it in the app: Kaleidohedra, 3D+ → OCD Stella Crystals.